

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Social media platforms have emerged as dominant channels for information dissemination, enabling users to rapidly share news content across geographical and cultural boundaries. While this connectivity facilitates information access, it has also accelerated the spread of fake news—misleading or false information presented as factual—which poses serious threats to public trust, democratic processes, and social stability. A defining characteristic of modern fake news is its multimodal nature, where textual claims are frequently accompanied by images that may be irrelevant, outdated, or intentionally misleading, thereby increasing the perceived credibility of false narratives.

Early approaches to fake news detection primarily relied on text-only analysis, employing classical machine learning algorithms such as Logistic Regression, Support Vector Machines, and Naïve Bayes using handcrafted linguistic features or TF-IDF representations. Subsequent deep learning-based methods introduced neural architectures, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and transformer-based models such as BERT, to capture richer semantic patterns in textual content. Although these methods achieved high accuracy on curated datasets, they often suffer from poor generalization, as they learn dataset-specific correlations rather than verifying the factual correctness of claims. Moreover, text-only models are inherently limited in their ability to detect visually driven misinformation.

To overcome these limitations, recent research has explored multimodal fake news detection, combining textual and visual features using deep neural networks such as CNN-based image encoders and transformer-based text encoders, followed by feature fusion mechanisms. While multimodal learning improves performance by capturing cross-modal correlations, most existing approaches remain fully supervised and correlation-driven, lacking explicit mechanisms for factual verification. As a result, these models may confidently misclassify claims when exposed to novel events, adversarial misinformation, or out-of-distribution content. Furthermore, many multimodal deep learning models operate as black boxes, offering

limited interpretability, which restricts their usability in real-world moderation and decision-support systems.

A critical shortcoming of existing approaches is the absence of external knowledge integration. Fake news often involves claims about real-world entities, events, or statistics that cannot be reliably assessed using internal dataset patterns alone. Without verification against trusted external sources, detection models cannot distinguish between factually supported and unsupported claims. Additionally, the lack of explainability raises concerns regarding transparency and trust, especially in high-stakes applications such as political misinformation monitoring and public health communication.

In this work, we propose an explainable multimodal fake news detection framework that integrates retrieval-augmented evidence verification, image–text consistency analysis, and supervised classification within a unified architecture. The proposed system processes textual content using feature-based or contextual text encoders (TF-IDF or Sentence-BERT) and evaluates visual–textual alignment using a pretrained vision–language model (CLIP). To incorporate factual grounding, the system employs a retrieval-augmented reasoning mechanism, where semantic queries derived from textual embeddings are used to retrieve relevant external evidence from trusted knowledge sources via FAISS-based vector similarity search. An evidence consistency score is computed to quantify the degree of alignment between the claim and retrieved factual information.

The extracted textual features, image–text consistency score, and evidence consistency score are then combined through a feature fusion strategy and passed to a supervised classification algorithm based on Logistic Regression, which learns the relative importance of each signal. This design choice ensures interpretability while enabling effective multimodal reasoning. The final output includes not only a fake or real prediction and confidence score, but also the retrieved evidence supporting the decision, thereby providing transparency and explainability.

1.2 MOTIVATION

The motivation for this project arises from the increasing sophistication of fake news on social media platforms. Modern misinformation often combines misleading text with manipulated or contextually unrelated images, making traditional detection methods insufficient. Although deep learning and multimodal approaches have improved classification accuracy, they rarely

verify claims against trusted external sources or explain the reasoning behind their predictions. Academically, this project is motivated by the need to integrate multiple research directions—multimodal learning, retrieval-based factual verification, and explainable artificial intelligence—into a unified framework. Practically, there is a strong demand for transparent and trustworthy fake news detection systems that not only classify content but also justify decisions using evidence and confidence measures.

Additionally, the availability of large-scale multimodal datasets and pretrained models provides an opportunity to design an efficient yet powerful system without training models from scratch. The project is further motivated by the goal of developing an interactive and deployable solution that can accept real-time user input and provide interpretable outputs, aligning with current industry and research trends.

1.3 SCOPE OF PROJECT

The design and deployment of a multimodal false news detection system that examines textual and visual content from social media posts are included in the project's scope. The system integrates text encoding, image–text consistency analysis, retrieval-augmented evidence verification, and supervised classification to assess whether a particular post is bogus or authentic.

The research also focuses on creating explainable outputs, including confidence scores, retrieved factual information, and human-readable reasoning. Using fresh user-provided inputs, an interactive interface is taken into consideration to showcase the system's capabilities. Due to time and budget limitations, the scope does not include multilingual identification, video-based misinformation analysis, large-scale real-time data crawling, or complete production-level deployment. Advanced social network graph modelling and continuous online learning are indicated as prospective future upgrades rather than part of the current implementation.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

Fake news identification has developed as crucial research subject due to the quick transmission of incorrect information on social media platforms and its enormous societal, political, and economic repercussions. The information modality (text-only or multimodal), the learning paradigm (supervised or semi-supervised), and the degree of reasoning and factual grounding used can all be used to broadly classify previous research.

The main goal of early research was to employ statistical and machine learning methods to find misleading linguistic patterns. Researchers started using deep learning models to identify semantic and contextual relationships in content as disinformation tactics changed. By using multimodal data, more recent research has expanded this field of study, recognising that fake news frequently blends textual assertions with deceptive visual cues to increase credibility.

Despite these developments, many present systems rely largely on correlation-based learning and fully labelled datasets, restricting their adaptability to emerging disinformation patterns. Additionally, a large amount of the literature is devoid of external factual verification procedures, which leaves such algorithms susceptible to confidently misclassifying new or unclear content. Most detection systems lack explainability, which further limits their usefulness in high-stakes situations where trust and transparency are crucial.

2.1.1 Traditional Text-Only Fake News Detection

Early research in fake news detection primarily relied on textual analysis using classical machine learning techniques. Several studies evaluated the effectiveness of traditional classifiers such as Logistic Regression, Support Vector Machines, Random Forests, and Naïve Bayes in combination with handcrafted linguistic features or TF-IDF representations. Comparative analyses demonstrated that while convolutional neural networks can outperform classical machine learning models on certain datasets, both approaches remain fully supervised and limited to textual cues [Benaouda et al., 2024]. Similar findings were reported in NLP-driven fake news detection studies that employed Bag-of-Words and TF-IDF features with

standard classifiers, achieving competitive accuracy while remaining sensitive to feature engineering choices and dataset-specific patterns [Coban and Bakal, 2025].

Additional investigations explored lightweight machine learning pipelines combining TF-IDF representations with linear and tree-based classifiers, highlighting their computational efficiency and ease of deployment [Gupta et al., 2022]. With the advancement of deep learning, hybrid architectures integrating contextual embeddings and sequential models were introduced to improve semantic understanding of news content. In particular, BERT-CNN-BiLSTM architectures demonstrated improved classification performance by capturing contextual and temporal dependencies within text [Alghamdi et al., 2022].

2.1.2 Multimodal Fake News Detection

To address the limitations of text-only methods, recent research has increasingly focused on multimodal fake news detection by jointly modeling textual and visual information. Multimodal frameworks integrating BERT-based textual encoders with convolutional neural network-based image feature extractors have been shown to improve detection performance compared to unimodal baselines by capturing complementary cross-modal signals [Reddy et al., 2025]. Further studies explored multimodal interaction mechanisms using unsupervised feature fusion strategies to model correlations between text and images more effectively [Geng et al., 2025].

Beyond simple fusion, cross-modal alignment techniques have been proposed to explicitly measure the semantic consistency between textual claims and associated images. Contrastive learning-based frameworks demonstrated that modeling image–text alignment can improve detection accuracy in cases where misleading visual content is used to reinforce false narratives [Wang et al., 2025]. Attention-based multimodal architectures combining transformer-based text encoders with CNN-based visual encoders further enhanced feature interaction by selectively focusing on relevant multimodal cues [Wang et al., 2024].

2.1.3 Semi-Supervised Fake News Detection

Due to the limited availability of labeled fake news data, several studies have investigated semi-supervised learning approaches to leverage large volumes of unlabeled content. Graph-based semi-supervised frameworks have been proposed to propagate labels across similarity graphs constructed using textual embeddings, demonstrating that competitive detection performance can be achieved with significantly fewer labeled samples [Guacho et al., 2018]. Subsequent

research extended this direction by employing convolutional neural networks within semi-supervised learning settings to exploit unlabeled social media data more effectively [Mansouri et al., 2020].

Recent studies leveraged CLIP-based representations to directly model image–text semantic alignment for fake news detection. CLIP-based frameworks demonstrated effectiveness in identifying inconsistencies between visual and textual content by computing similarity scores between modalities [Zhou et al., 2022]. More recent studies combined semi-supervised learning with graph neural networks to propagate credibility labels across social or content-based graphs, showing improved robustness under limited supervision [Benamira et al., 2019].

2.1.4 Graph-Based and Propagation-Oriented Detection

Another important line of research focuses on modeling the propagation dynamics of fake news using graph neural networks. By leveraging social interaction networks, repost structures, and diffusion patterns, these approaches aim to distinguish fake news from real news based on how information spreads across social platforms. Empirical studies have demonstrated that propagation-based representations captured using graph neural networks can effectively identify misinformation in social networks [Mishra et al., 2025].

However, propagation-oriented methods often rely on weak or shallow content representations and do not explicitly verify the factual correctness of claims. In addition, their effectiveness is limited when propagation data is incomplete or unavailable, motivating the need for content-aware and evidence-grounded detection frameworks.

2.1.5 LLM-Based Fake News Detection Systems

Recent advances in large language models have led to increased interest in prompt-based fake news detection and information reliability assessment. Several frameworks integrate LLM reasoning with retrieval-based evidence to improve the interpretability and trustworthiness of misinformation detection systems [Turaga et al., 2024]. Retrieval-augmented LLM pipelines have also been explored for claim verification, combining semantic retrieval with generative inference to assess content credibility [Bai and Fu, 2024].

2.1.6 Retrieval-Augmented Fact-Checking Systems

Retrieval-Augmented Generation has been widely adopted to mitigate hallucination in neural reasoning systems by grounding predictions in external factual knowledge. Recent studies demonstrate that integrating retrieval mechanisms with generative models improves claim verification accuracy by explicitly linking predictions to supporting evidence [ACM Workshop, 2024]. However, most retrieval-augmented systems are designed for fact-checking or question-answering tasks rather than end-to-end fake news classification and remain largely limited to textual inputs.

2.2 GAPS IDENTIFIED

- A significant portion of existing fake news detection approaches rely solely on textual information, ignoring visual content that often plays a critical role in spreading misinformation on social media platforms.
- While multimodal methods incorporate both text and images, most of them focus on learning correlations between modalities rather than verifying the factual correctness of the claims being made.
- Many multimodal and deep learning-based models operate as black-box classifiers, providing predictions without transparent explanations or justifications that can be understood by human users.
- Existing approaches rarely integrate external factual knowledge sources, such as encyclopaedic databases or fact-checking repositories, to support or refute the claims present in social media content.
- Semi-supervised learning techniques demonstrate the potential to utilize unlabelled data but are predominantly limited to text-only representations and do not effectively leverage multimodal information.
- Propagation-based and graph neural network approaches capture the spread and virality of fake news but often neglect the semantic content and factual validity of the information itself.
- Large language model-based detection systems enhance reasoning capability but are prone to hallucinations when used without structured retrieval and evidence grounding mechanisms.

- Most retrieval-augmented systems are designed for claim verification or question-answering tasks and do not perform end-to-end fake news classification in a multimodal setting.
- Several models achieve high accuracy on curated or domain-specific datasets but exhibit limited generalization when applied to real-world, noisy, or evolving social media content.

2.3 OBJECTIVES

The primary objective of this project is to design and implement a robust and explainable fake news detection framework capable of handling multimodal social media content. The system aims to move beyond traditional pattern-based classification by incorporating external factual evidence, multimodal consistency analysis, and transparent reasoning to improve reliability in real-world misinformation scenarios.

In order to achieve this goal, the project focuses on the following specific objectives:

- Analyze social media posts by jointly processing textual content and associated visual information to detect misleading cross-modal relationships.
- demonstrate effective extraction of textual representations using embedding techniques and visual consistency scores using image–text alignment models.
- Integrate a retrieval-augmented evidence verification mechanism that retrieves relevant external knowledge from trusted sources to support or refute claims.
- Compute an evidence consistency score that quantitatively reflects the factual alignment between a claim and retrieved external information.
- Fuse multimodal features, including textual features, visual consistency scores, and evidence consistency scores, into a unified representation for classification.
- Develop a supervised classification model capable of accurately distinguishing between fake and real news along with confidence scores.
- Generate explainable outputs that provide human-readable reasoning and supporting evidence to enhance interpretability and user trust.
- Design the framework in a modular and extensible manner to allow future enhancements such as advanced deep learning models, social propagation analysis, or real-time deployment.

2.4 PROBLEM STATEMENT

The way we share news online has changed everything, but it's also made it way too easy for fake news to spread. Nowadays, misinformation isn't just a lie in a text post; it often mixes misleading words with edited or out-of-context photos to trick us. This is a huge problem because it ruins public trust and can even cause chaos in real life. Most of the tools we have right now aren't good enough to stop it. Some only look at the words and ignore the pictures, while others just look for patterns but don't actually check if the facts are true. Even the newest AI models sometimes "hallucinate," making up their own facts instead of looking for real evidence.

Another big issue is that most detection systems are like a "black box"—they tell you something is fake but never explain “Why”. This makes them hard to use for people like journalists or government officials who need to see the proof. Because fake news is always changing and we don't have enough labeled data to train perfect systems, we need a better approach. This project introduces a new, smarter system that looks at both text and images at the same time. It doesn't just guess; it actually goes out, finds real-world evidence, and explains its reasoning in a way that humans can understand. The goal is to make social media a safer, more honest place by providing a tool that is reliable, clear, and ready for the real world.

Social media makes it incredibly easy for fake news to spread by combining misleading stories with edited images. For example, someone might share a real photo of a natural disaster but claim it is happening right now in a different city to cause panic. Current AI tools often miss this because they might check the text but ignore the photo, or they might flag it as "fake" without explaining why. This project creates a system that acts like a digital detective; it would see the photo, find the original source online, and show you that it actually from five years ago. By providing this specific evidence, the system helps people understand exactly how they are being misled.

2.5 PROJECT PLAN

The proposed project is executed in a structured, stage-wise manner to ensure systematic development, validation, and evaluation of the multimodal fake news detection framework. The project plan is designed to progressively enhance the system’s capability, starting from fundamental data preparation and moving toward evidence-grounded reasoning and user-facing deployment. Each stage builds upon the outcomes of the previous phase, enabling controlled experimentation and performance analysis throughout the project lifecycle.

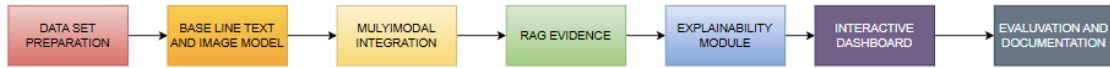


Fig 2.5: Flowchart of Project Execution

The project begins with dataset preparation, where multimodal social media data consisting of textual content, associated images, and class labels are collected and curated. This stage involves cleaning noisy textual inputs, filtering unusable or broken image links, handling missing values, and selecting representative subsets of data for experimentation. The objective of this phase is to ensure data reliability and consistency before model development, while also addressing practical challenges such as incomplete visual data commonly encountered in real-world social media datasets.

Following data preparation, a baseline text and image modeling phase is conducted to establish initial performance benchmarks. Textual content is processed using conventional text representations, while visual features are extracted independently using pretrained vision models. This phase allows the evaluation of unimodal and loosely coupled multimodal features, providing insight into the individual contribution of text and image modalities to fake news detection. The baseline results serve as a reference point for assessing improvements introduced by subsequent stages.

The next phase focuses on multimodal integration, where textual embeddings and image–text consistency scores are jointly modeled. By integrating semantic textual features with visual–textual alignment signals, the system moves beyond isolated modality analysis and begins to

capture cross-modal inconsistencies that frequently characterize deceptive content. This stage emphasizes feature fusion and joint reasoning rather than deep end-to-end learning, maintaining interpretability and computational efficiency.

To strengthen factual reliability, the project incorporates a Retrieval-Augmented Generation (RAG)-based evidence verification stage. In this phase, textual representations are used to retrieve semantically relevant information from trusted external knowledge sources using FAISS-based vector similarity search. Retrieved evidence is evaluated for semantic consistency with the original claim, producing an evidence consistency score that reflects factual support or contradiction. This stage transforms the system from a correlation-based classifier into an evidence-aware detection framework.

Subsequently, an explainability module is introduced to enhance transparency and user trust. Rather than influencing the prediction process, a large language model is employed exclusively to generate human-readable explanations grounded in retrieved evidence, multimodal consistency scores, and classifier confidence values. This separation ensures that explanations remain factual, interpretable, and resistant to hallucination, addressing a key limitation of black-box deep learning models.

The system is then integrated into an interactive dashboard, implemented using lightweight web-based frameworks such as Gradio. This interface enables users to input new social media posts in the form of text and images and receive real-time predictions, confidence scores, retrieved evidence, and natural-language explanations. The dashboard serves as both a demonstration platform and a practical deployment interface, bridging the gap between research experimentation and real-world usability.

Finally, the project concludes with evaluation and documentation, where model performance is systematically analyzed using standard metrics such as accuracy, precision, recall, and F1-score. Qualitative assessments of retrieved evidence and explanation quality are also conducted to validate interpretability. The final phase involves documenting experimental findings, limitations, and future extensions, ensuring reproducibility and readiness for academic review and publication.

CHAPTER 3

TECHINICAL SPECIFICATION

3.1 FEASIBILITY STUDY

3.1.1 Technical Feasibility

The proposed multimodal fake news detection system is technically feasible as it relies entirely on open-source tools, libraries, and pretrained models. The framework is implemented using widely adopted machine learning and deep learning libraries such as PyTorch, HuggingFace Transformers, Scikit-learn, FAISS, and Sentence-Transformers. The use of pretrained models such as CLIP and Sentence-BERT eliminates the need for training large models from scratch, thereby reducing computational complexity. The system is modular in nature and can be executed on standard computing environments, making it suitable for academic research and scalable experimentation.

3.1.2 Social Feasibility

Fake news poses serious social challenges by influencing public opinion, spreading misinformation, and creating panic during critical events. The proposed system contributes positively to society by providing an automated and explainable solution for identifying fake news on social media platforms. By offering evidence-based explanations along with predictions, the system improves transparency and user trust. The framework can assist journalists, researchers, and social media users in verifying information, thereby supporting responsible information consumption and promoting digital literacy.

3.2 SYSTEM SPECIFICATION

3.2.1 Software Specification

- **Google Colab:** Used as the primary development and experimentation environment for implementing, testing, and visualizing the multimodal fake news detection pipeline.
- **PyTorch:** Utilized for loading and executing pretrained deep learning models, particularly the CLIP model, and for efficient tensor-based computations.

- **Scikit-learn:** Employed for feature engineering, dataset splitting, training the Logistic Regression classifier, and evaluating model performance using standard metrics.
- **HuggingFace Transformers:** Used to access pretrained transformer-based models and utilities, enabling seamless integration of state-of-the-art NLP and multimodal components.
- **Sentence-Transformers:** Applied for generating semantic sentence-level embeddings required for evidence retrieval and similarity-based reasoning.
- **CLIP (Contrastive Language–Image Pretraining):** Used to compute image–text consistency scores by projecting visual and textual inputs into a shared embedding space.
- **FAISS (Facebook AI Similarity Search):** Implemented for efficient indexing and retrieval of external factual evidence using vector similarity search.
- **Pandas:** Used for dataset loading, preprocessing, filtering, and structured data manipulation.
- **NumPy:** Utilized for numerical operations, feature fusion, and efficient array-based computations.
- **IPython Display / Gradio (Optional):** Used for visualizing images, displaying predictions, and presenting retrieved evidence and explanations in an interactive format.

3.2.2 Hardware Specification

The proposed system does not require specialized hardware components, as it is designed as a software-centric research framework. The implementation and experimentation are carried out on general-purpose computing platforms such as personal computers or cloud-based environments. Optional GPU support is utilized when available to accelerate model inference; however, the system remains functional on CPU-only environments, making it accessible for academic and research use.

CHAPTER 4

DESIGN APPROACH AND DETAILS

4.1 SYSTEM ARCHITECTURE

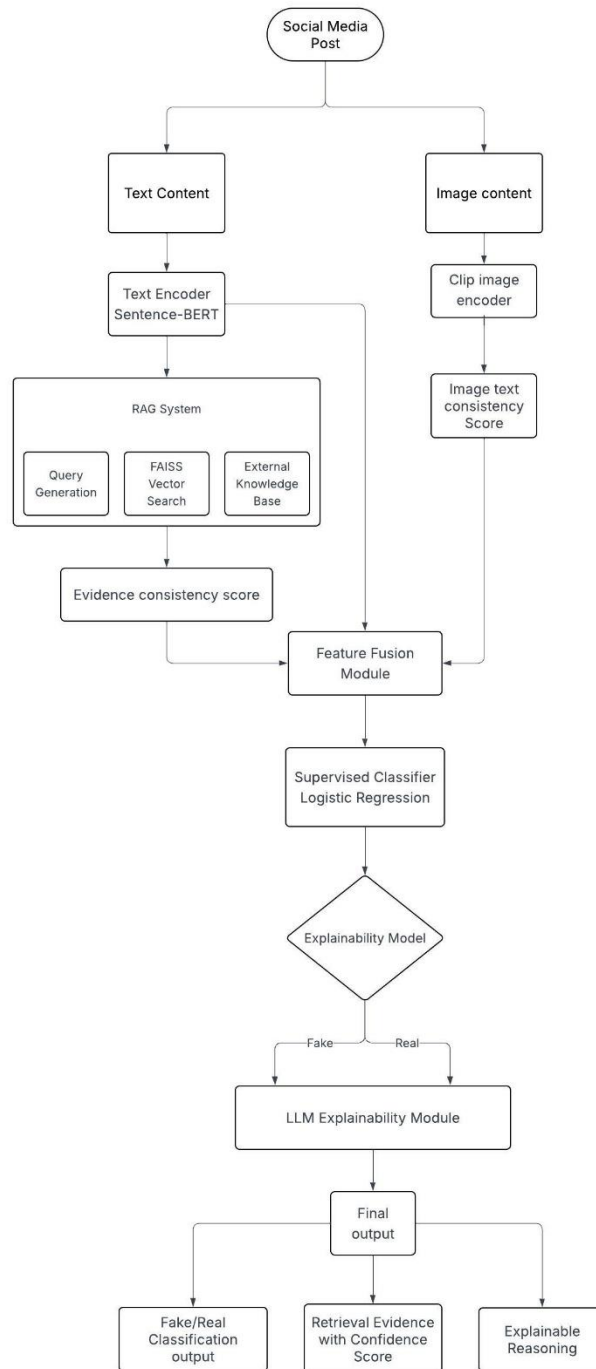


Fig 4.1: Proposed Architecture Diagram for Fake news detection

At the input layer, the system accepts a social media post as a composite entity containing textual and visual components. Instead of merging modalities at an early stage, the architecture adopts a parallel processing strategy in which text and image content are handled independently during initial analysis. This design choice allows each modality to be processed using representations best suited to its semantic characteristics. The textual branch of the architecture employs a Sentence-BERT-based text encoder to transform raw text into dense semantic embeddings. These embeddings provide a high-level representation of the claim expressed in the post, capturing contextual meaning rather than relying on surface-level lexical patterns. The encoded textual representation serves as a central reference point for downstream verification and integration modules. To support factual validation, the architecture incorporates a Retrieval-Augmented Generation (RAG) subsystem as a dedicated evidence verification layer. This subsystem encapsulates internal mechanisms for semantic query formulation, FAISS-based vector similarity search, and interaction with external knowledge sources such as Wikipedia and fact-checking repositories. Rather than producing a generative response, the RAG system outputs structured evidence and a numerical evidence consistency score that reflects the degree of alignment between the post’s claim and retrieved factual information. This separation ensures that factual grounding remains explicit and quantifiable within the architecture.

In parallel with textual processing, the visual branch of the architecture analyzes image content using a CLIP-based image encoder. CLIP enables both textual and visual representations to be projected into a shared semantic space, allowing the architecture to assess the coherence between an image and its accompanying text. The resulting image-text consistency score provides an important signal for identifying visually misleading content, such as images reused outside their original context. The outputs from the text encoder, the RAG-based evidence verification layer, and the visual consistency module converge at the feature fusion module. This component is responsible for aggregating heterogeneous information into a unified feature representation. By combining semantic, factual, and cross-modal signals, the architecture ensures that classification decisions are informed by complementary perspectives rather than isolated cues.

The fused feature representation is subsequently passed to a supervised classification layer implemented using Logistic Regression. Within the architectural context, this classifier functions as a transparent decision-making component rather than a deep black-box model. Its role is to map evidence-enriched multimodal features to credibility labels while preserving interpretability and stable decision boundaries.

To address the requirement for transparency and user trust, the architecture integrates an explainability layer positioned after the classification stage. A decision logic component determines the predicted credibility class, which is then forwarded to an LLM-based explainability module. Importantly, the large language model is architecturally isolated from the prediction pipeline and is utilized solely for generating human-readable explanations. It synthesizes retrieved evidence, consistency scores, and classifier confidence values to produce grounded explanations that justify the final decision without influencing the classification outcome itself. The final output layer presents the system's results in a structured form, including the fake or real classification label, confidence score, retrieved supporting evidence, and an explanatory narrative. This layered architectural design ensures that the system not only detects misinformation but also communicates the rationale behind its decisions in an interpretable and evidence-driven manner.

4.2 DESIGN

4.2.1 Data Flow Diagram

- This Data flow diagram shows how a social media post containing both text and image is taken as input and processed through different modules such as text processing, image processing, and evidence retrieval in a sequential and logical manner.
- It clearly illustrates the parallel flow of information, where image-text consistency is checked using CLIP while factual verification is performed through evidence retrieval and RAG reasoning, ensuring both semantic and factual validation.
- The diagram highlights how all extracted and verified information is finally fused to generate a fake or real prediction along with an explainable output, which is then presented back to the user in an understandable form.

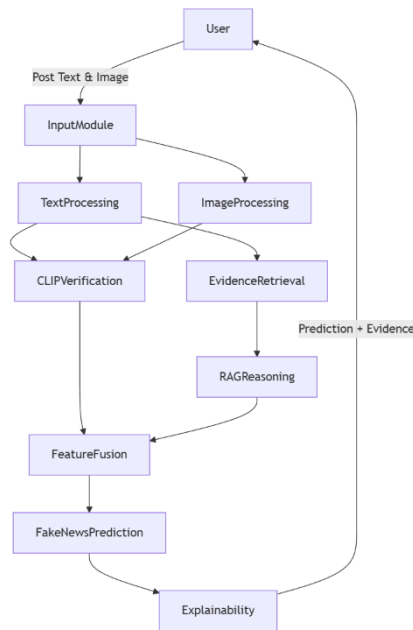


Fig 4.2.1: Dataflow Diagram of this Project

4.2.2 Class Diagram

- This class diagram represents the structural design of the system by organizing the fake news detection pipeline into well-defined classes such as text processing, image processing, evidence retrieval, classification, and explainability.
- It shows how textual and visual features are processed independently and later combined through the CLIP model and feature fusion module, ensuring clear separation of responsibilities among system components.
- The diagram highlights the interaction between decision-making and explainability, where the final classifier output is passed to the explainability module and dashboard to provide both prediction results and supporting evidence to the user.

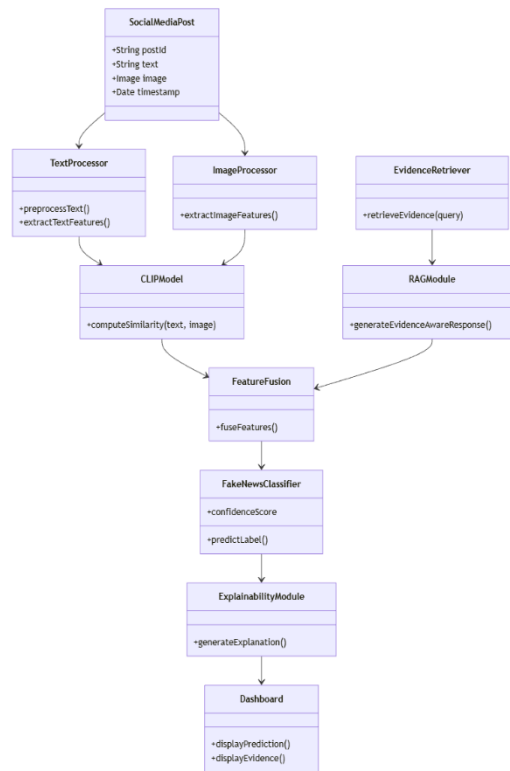


Fig 4.2.2: Class Diagram of the Project

CHAPTER 5

METHODOLOGY AND TESTING

5.1 MODULE DESCRIPTION

The proposed framework presents a unified multimodal fake news detection methodology that integrates semantic content understanding, visual–textual consistency analysis, external evidence verification, and explainable decision-making. Unlike conventional approaches that rely on isolated modalities or purely correlation-based learning, the proposed system emphasizes factual grounding and interpretability while maintaining computational efficiency.

- A unified multimodal framework is designed to detect fake news by jointly analyzing textual content, visual information, and external factual evidence, with a strong focus on interpretability and reliability.
- Textual claims from social media posts are encoded using Sentence-BERT to generate semantic embeddings that capture contextual meaning and support robust similarity-based reasoning.
- Factual verification is performed using a Retrieval-Augmented Generation (RAG) mechanism, where relevant evidence is retrieved from trusted knowledge sources through FAISS-based vector search and compared with the original claim.
- Visual content is analyzed using the CLIP model to compute image–text consistency scores, enabling the detection of misleading or contextually irrelevant images associated with false narratives.
- Multimodal information, including text embeddings, evidence consistency scores, and image–text similarity scores, is combined through a feature fusion module to form a comprehensive representation of the news content.
- A supervised Logistic Regression classifier operates on the fused features to generate fake or real predictions along with interpretable confidence scores.
- An explainability module utilizes a large language model to produce evidence-grounded, human-readable explanations that clearly justify the model’s predictions.

- The final system output includes the classification label, confidence score, retrieved supporting evidence, and a concise explanation, ensuring transparency and practical usability in real-world scenarios.

5.2 TESTING:

5.2.1 Data Collection and Preprocessing:

The availability of large-scale multimodal datasets has played a crucial role in advancing fake news detection research. The Fakeddit dataset provides a comprehensive benchmark containing fine-grained fake news labels along with associated visual content, enabling systematic evaluation of unimodal and multimodal models [Nakamura et al., 2020]. While this dataset has facilitated significant progress in multimodal detection, the baseline models provided do not incorporate external evidence retrieval, reasoning capabilities, or explainability mechanisms, highlighting opportunities for further methodological development. Each data instance consists of a cleaned news headline (`clean_title`), a corresponding image URL, and a two-class label (`2_way_label`) indicating the veracity of the content.

In the initial phase of experimentation, the raw dataset was filtered to retain only samples containing both textual content and valid image URLs, ensuring the availability of multimodal inputs for analysis. Records with missing text, missing labels, or inaccessible image URLs were removed. This preprocessing step is necessary to guarantee consistency across the text and image processing pipelines and to avoid introducing noise caused by incomplete multimodal data. To enable efficient experimentation and rapid prototyping, a subset of 1,000 samples was randomly selected from the cleaned dataset during the current development phase. From this subset, only samples with successfully downloaded images were retained, resulting in a reduced but reliable working set of usable multimodal instances. This staged sampling strategy is intentionally adopted to balance computational feasibility and methodological validation during early experimentation.

Text preprocessing involves lightweight normalization consistent with the dataset’s pre-cleaned format. The textual content is vectorized using the Term Frequency–Inverse Document Frequency (TF-IDF) scheme, which transforms each news title into a fixed-dimensional

numerical representation while preserving discriminative lexical patterns. The TF-IDF vocabulary size is capped at 3,000 features to reduce sparsity and computational overhead. For visual preprocessing, images are retrieved directly from their URLs and converted into RGB format. Images that fail to load within a predefined timeout threshold are discarded to maintain robustness. Visual features are not handcrafted; instead, semantic representations are extracted using a pretrained CLIP image encoder, which enables direct comparison between textual and visual modalities. External factual knowledge is incorporated through retrieval-based preprocessing. For a subset of textual inputs, concise evidence snippets are collected from publicly available knowledge sources such as Wikipedia. These external documents are embedded using a sentence-level transformer model and indexed using FAISS for efficient similarity-based retrieval. The resulting evidence embeddings are used to compute an evidence consistency score, which quantifies the semantic alignment between a claim and retrieved factual information.

In the final system configuration, the feature space for each sample consists of three components: TF-IDF textual features, CLIP-based image–text consistency scores, and retrieval-based evidence consistency scores. While the current experimental results are based on a limited subset of the dataset, the same preprocessing pipeline is designed to scale to larger sample sizes (20,000–50,000 instances) in the final evaluation phase, subject to computational constraints.

5.2.2 Evaluation Results:

This section reports the results obtained up to the current development stage, corresponding to the Phase-1 baseline and Phase-2 evidence-augmented model.

In the baseline multimodal setting, TF-IDF text features are combined with CLIP-based image–text consistency scores and used to train a supervised Logistic Regression classifier. The dataset is split into training and validation sets using an 80:20 stratified split to preserve class balance. The baseline model achieves a validation accuracy of approximately 70%, indicating that simple multimodal feature fusion already provides meaningful discrimination between fake and real news.

In the second phase, external factual evidence is incorporated through a retrieval-augmented mechanism. For each input text, relevant evidence is retrieved from an external knowledge corpus using FAISS-based similarity search. An evidence consistency score is computed using

cosine similarity between the claim embedding and retrieved evidence embeddings. This score is appended to the multimodal feature vector and fed into the classifier.

The evidence-augmented model maintains a validation accuracy of approximately 70%, while significantly improving qualitative interpretability. Unlike purely classification-driven models, the proposed framework provides retrieved evidence snippets, confidence scores, and human-readable explanations that justify predictions. Sample outputs demonstrate the system’s ability to identify weak or absent factual support for misleading claims, even when visual and textual content appear superficially plausible.

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*** Validation Accuracy (External RAG): 0.7
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Classification Report:

	precision	recall	f1-score	support
0	0.73	0.52	0.60	62
1	0.69	0.85	0.76	78
accuracy			0.70	140
macro avg	0.71	0.68	0.68	140
weighted avg	0.71	0.70	0.69	140

Fig 5.2.2 Evaluation results of Classification through CLIP usage till now

Some Examples from Dataset for Output:

Input Text:
president obama meets with presidentelect trump

Associated Image:

Prediction:
FAKE (Confidence: 0.76)

Retrieved Evidence:
1. The president of the United States (POTUS) is the head of state and head of government of the United States. The president directs the executive branch of the federal government.
2. The United States and Russia maintain strategic foreign relations. They have had diplomatic relations since the establishment of the latter country in 1991, a continuation of the relationship between the United States and the Soviet Union.

Explanation:
The claim has weak or no support from external factual sources.

Image-Text Consistency Score (CLIP): 32.36

Fig 5.2.2 use case of the data with both text and image combined output of detecting Fake/Real



Fig 5.2.2 Same output on detecting the Fake/Real with gathered evidence upto now

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